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To cite this article: Rohan Kumar Choudhary and Amit Kumar Jha 2025 *IOP Conf. Ser.: Mater. Sci. Eng.* **1333** 012002

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## Structural Assessment and Rehabilitation of High-Rise Buildings: A Case Study of RNA Mirage Tower Mumbai

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### Abstract

The rapid pace of urbanization and the rising demand for space in metropolitan areas have led to a surge in the construction of high-rise buildings. However, the long-term performance and durability of these structures often suffer due to inadequate construction practices and poor maintenance. This study presents a comprehensive case analysis of the RNA Mirage building, a G+40 residential tower located in Worli, Mumbai, which exhibited significant structural distress after 15 years of service. Visible signs of deterioration included widespread spalling, reinforcement corrosion, water leakage, and cracking in both structural and non-structural elements. A detailed investigation was conducted using a combination of visual inspections, non-destructive testing (NDT), and semi-destructive techniques. These included the Rebound Hammer Test, Ultrasonic Pulse Velocity Test, Half-Cell Potential Test, Carbonation Depth Test, and Core Sampling. The results indicated considerable variation in the compressive strength of concrete, with some readings as low as 25 N/mm<sup>2</sup>, highlighting poor original construction quality and increased permeability. This facilitated the ingress of harmful agents such as chlorides and carbon dioxide, leading to corrosion of steel reinforcement and concrete degradation. Based on the assessment, a structured rehabilitation process was implemented, including the use of polymer-modified mortar, corrosion inhibitors, guniting, and waterproofing treatments. This case underscores the critical need for quality control during construction and the importance of periodic maintenance in preserving the safety and longevity of high-rise structures. Key recommendations are provided to improve durability in future projects and minimize the need for extensive structural rehabilitation.

**Keywords:** High-Rise Buildings, Structural Rehabilitation, Corrosion, Non-Destructive Testing (NDT), Structural Health Monitoring



## Introduction

The increasing demand for high-rise structures is a direct consequence of rapid urbanization and the scarcity of available land in metropolitan regions. As land prices continue to surge—particularly in Tier-1 cities—vertical development has emerged as a practical and efficient solution to accommodate the growing urban population. However, while the construction of high-rise buildings is on the rise, such projects are inherently time-intensive and must navigate through numerous regulatory approvals. [1-4]. The long-term performance and service life of these structures are critical considerations. Unfortunately, poor construction practices and inadequate maintenance can jeopardize structural integrity, leading to premature deterioration and escalating repair costs. Maintenance, therefore, plays a pivotal role in ensuring the durability and operational efficiency of high-rise buildings. It not only prevents structural degradation but also enhances the safety, comfort, and overall well-being of residents and visitors. Through regular and effective maintenance, both the functional and aesthetic values of buildings can be preserved over time. [1-6].

In urban settings—especially in Tier-1 cities—the maintenance of high-rise buildings presents several unique challenges. These include limited accessibility, environmental exposure, and high occupancy levels, all of which necessitate timely inspections and proactive interventions to identify and mitigate early signs of distress.

Additionally, the natural aging of buildings gradually diminishes their functional performance. In response, modern structural health monitoring technologies have evolved significantly, offering improved assessment capabilities and enabling more proactive maintenance strategies. These tools support informed decision-making and bolster the durability and efficiency of maintenance practices. [2-8].

Globally, maintenance is recognized as a crucial contributor to economic growth. For instance, in countries like the USA, Hong Kong, and Saudi Arabia, maintenance-related expenditures account for nearly 50% of the construction industry's total turnover and represent approximately two-thirds of contractual agreements. This highlights the dual importance of maintenance—from both a structural and economic perspective. [6]

Moreover, studies indicate that design deficiencies account for 50% to 60% of long-term maintenance costs. This underscores the necessity of thorough planning and

thoughtful design during the early stages of construction to minimize future repair demands.[8]

To ensure effective maintenance, a strategic and well-coordinated approach is essential. This includes careful planning, accurate scheduling, adequate budgeting, and close collaboration among stakeholders. Skilled personnel, appropriate tools and equipment, and the use of high-quality materials further ensure the success of maintenance efforts.[2-4]

To better understand the challenges associated with maintaining high-rise structures, a case study was conducted on the RNA Mirage building—a G+40 residential tower located in Worli, Mumbai, as shown in Figure 1.



Figure-1- RNA Mirage building, a G+40 residential tower located in Worli, Mumbai.

### **Condition Assessment of the RNA Mirage Building**

The RNA Mirage building exhibited extensive structural distress, primarily attributed to reinforcement corrosion. Notable signs of deterioration included widespread spalling and delamination at column corners across various floors, as well as damage to the

soffits of beams, slabs, and architectural fins. Additional issues observed included leached plaster on parapets, corroded and leaking water and drainage pipes, and water stagnation on the terrace due to an improperly designed slope.

Structural cracks were also found in masonry infill walls, with certain areas showing complete disintegration, as illustrated in **Figure 2**. Severe debonding and degradation of plaster on both interior and exterior surfaces further emphasized the building's deteriorated condition.

Given the advanced state of deterioration and associated safety hazards, immediate rehabilitation was deemed necessary. This case study highlights the vital role of early intervention and regular maintenance in preserving the safety, usability, and lifespan of high-rise structures.



Figure-2- Structural defects and visible discrepancies and signs of distress

### **Repair and Rehabilitation Process**

The rehabilitation of the RNA Mirage building began with a preliminary visual inspection and a detailed review of the building's architectural drawings. Major structural defects were particularly noted on floors 1 and 4, as illustrated in **Figure 3**. A preliminary report was generated, outlining observed deficiencies and recommending further testing.



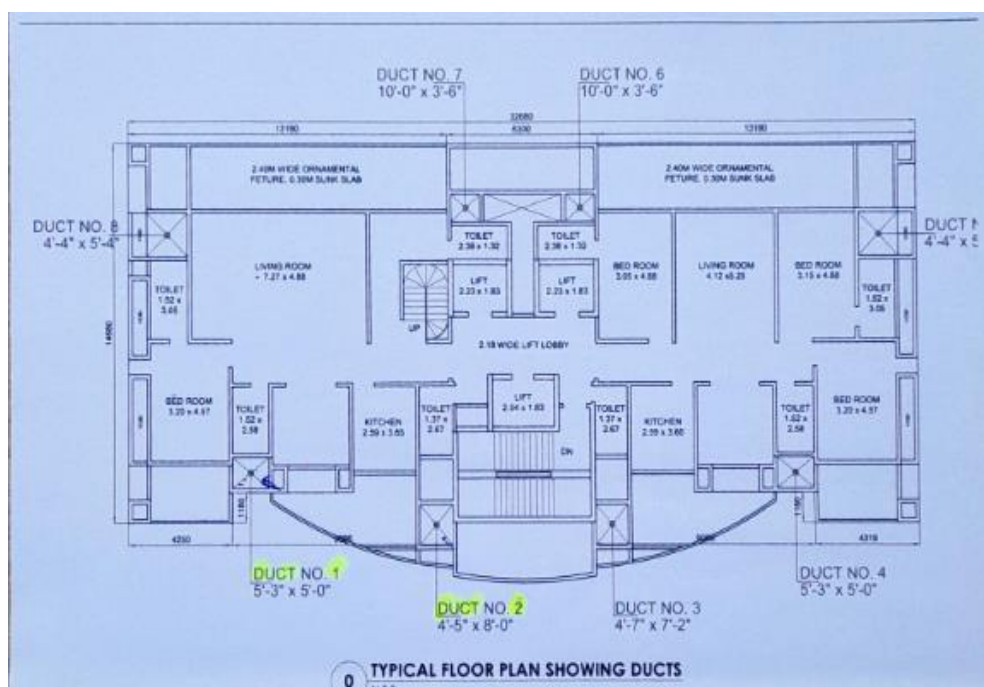
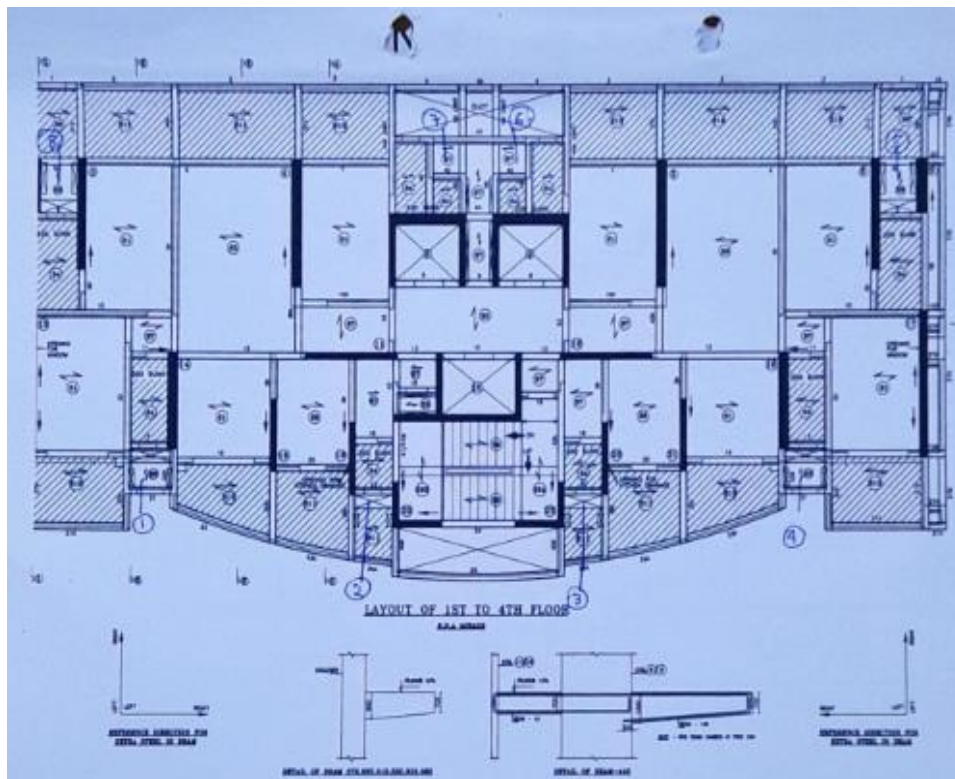


Figure-3- The floor plan and drawing of floor 1 and 4

A combination of non-destructive and semi-destructive testing was then conducted to evaluate the building's structural health. These tests assessed concrete quality, reinforcement condition, and moisture content. Following the testing phase, a comprehensive structural report was prepared, providing detailed analysis and recommended methodologies for repair.

The repair work commenced with the removal of deteriorated plaster and exposure of corroded reinforcement. The steel bars were cleaned using wire brushes, treated with rust remover, and coated with Nitocoat AP35 polymer to prevent future corrosion. Where necessary, additional reinforcement was introduced by welding new bars or anchoring with Hilti grouting techniques to ensure proper load transfer.

A polymer-modified mortar mix (1:5:15 ratio of AR Std polymer, cement, and quartz sand) was applied to the treated surfaces. This was followed by two layers of cement plaster. Severely deteriorated blockwork areas underwent guniting to restore structural strength. Post-guniting, two coats of waterproof plaster were applied for enhanced durability and moisture resistance, as shown in **Figure 4** (a) and (b) and the surface was finished with a water-based acrylic paint

For external wall waterproofing, a combination of Damp Seal and Seal-O Primer was used, offering long-term protection against moisture ingress. These interventions not only restored the building's structural integrity but also significantly improved its appearance and extended its service life.



**Figure 4** (a) - Deteriorated blockwork



**Figure 4 (b)** The surface was finished with a water-based acrylic paint

### Testing and Structural Assessment of Building

As part of the structural assessment and rehabilitation planning for the RNA Mirage Co-operative Housing Society, Worli, a series of non-destructive and semi-destructive tests were conducted. These tests were recommended by the Author and the appointed Structural Engineer to evaluate the current condition of the building and to determine the extent of deterioration. The results from each test are detailed below, presented in ascending order for clarity.

#### 1. Rebound Hammer Test

- **Project Name & Address:** RNA Mirage Co-op Housing Society, Worli, Mumbai
- **Type of Sample:** Concrete
- **Quantity:** 7 Nos
- **Condition:** Acceptable
- **Specification Followed:** IS 516 (Part 5/Sec 4): 2020
- **Testing Method:** Schmidt Rebound Hammer
- **Environmental Conditions:**



- Temperature: 32°C
- Relative Humidity: Ambient

**Results Summary:**

| Sr. No | Location/ID | Member | Hammer Position | Avg. Rebound No. | Avg. Compressive Strength (MPa) |
|--------|-------------|--------|-----------------|------------------|---------------------------------|
| 1      | KL6-A1      | Column | Horizontal      | 16               | 30                              |
| 2      | KL6-A2      | Column | Horizontal      | 15               | 29                              |
| 3      | KL6-A3      | Column | Horizontal      | 14               | 28                              |
| 4      | KL6-A4      | Column | Horizontal      | 25               | 33                              |
| 5      | KL6-A5      | Column | Horizontal      | 31               | 34                              |
| 6      | KL6-A6      | Column | Horizontal      | 20               | 30                              |
| 7      | KL6-A7      | Column | Horizontal      | 25               | 32                              |

**Observation:** The test revealed that the structural concrete in the columns retains satisfactory compressive strength, indicating no significant deterioration despite the building being 15 years old.

**2. Ultrasonic Pulse Velocity (UPV) Test**

- **Specification Followed:** IS 516 (Part 5/Sec 1): 2018
- **Environmental Conditions:**
  - Temperature: 32°C
  - Relative Humidity: Ambient

**Results Summary:**

| Sr. No | Location/ID | Member | Method   | Longitudinal Pulse Velocity (km/sec) |
|--------|-------------|--------|----------|--------------------------------------|
| 1      | KL6-A1      | Column | Indirect | 3.75                                 |
| 2      | KL6-A2      | Column | Indirect | 3.90                                 |

**Sr. No Location/ID Member Method Longitudinal Pulse Velocity (km/sec)**

|   |        |        |          |      |
|---|--------|--------|----------|------|
| 3 | KL6-A3 | Column | Indirect | 4.00 |
| 4 | KL6-A4 | Column | Indirect | 3.80 |
| 5 | KL6-A5 | Column | Indirect | 4.10 |
| 6 | KL6-A6 | Column | Indirect | 3.85 |
| 7 | KL6-A7 | Column | Indirect | 4.10 |

**Concrete Quality Grading:**

- Above 4.40 km/sec: Excellent
- 3.75 to 4.40 km/sec: Good
- 3.00 to 3.75 km/sec: Doubtful
- Below 3.00 km/sec: Poor

**Observation:** The results fall within the range of 3.75 to 4.10 km/sec, indicating that the overall concrete quality is good.

**3. Half-Cell Potentiometer Test**

- **Specification Followed:** ASTM C876: 2015

**Results Summary:**

| Sr. No | Location/ID | Member | Readings (mV)          | Avg (mV) |
|--------|-------------|--------|------------------------|----------|
| 1      | KL6-A1      | Column | -180, -191, -210, -170 | -188     |
| 2      | KL6-A2      | Beam   | -360, -290, -370, -280 | -325     |
| 3      | KL6-A3      | Column | -180, -260, -190, -210 | -210     |
| 4      | KL6-A4      | Beam   | -125, -255, -158, -166 | -176     |
| 5      | KL6-A5      | Column | -120, -190, -170, -120 | -150     |
| 6      | KL6-A6      | Beam   | -130, -170, -120, -190 | -153     |

**Interpretation:**

- More negative than -350 mV: High probability of corrosion
- -200 to -350 mV: Uncertain corrosion probability
- More positive than -200 mV: Low probability of corrosion

**Observation:** Most readings fall within the range indicating **uncertain to low probability of corrosion**, suggesting localized mild corrosion activity.

**4. Carbonation Depth Test**

- **Specification Followed:** IS 516 (Part 5/Sec 4): 2020

**Results Summary:**

| Sr. No | Location/ID | Member | Depth of Carbonation (mm) | Remarks                                 |
|--------|-------------|--------|---------------------------|-----------------------------------------|
| 1      | KL6-A1      | Column | 5                         | Slight carbonation, surface not changed |
| 2      | KL6-A2      | Column | 6                         |                                         |
| 3      | KL6-A3      | Beam   | 4                         |                                         |
| 4      | KL6-A4      | Column | 6                         |                                         |
| 5      | KL6-A5      | Column | 6                         |                                         |
| 6      | KL6-A6      | Column | 5                         |                                         |

**Observation:** The presence of carbonation was detected up to 6 mm depth, indicating mild corrosion due to CO<sub>2</sub> ingress, though not structurally alarming at this stage.

**5. Steel Rebar Tensile Strength Test**

**Observation:** The tensile strength of the reinforcement bars showed some degradation due to corrosion and chloride attack. While the reduction in strength is not critical, it necessitates protective and restorative measures.

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## 6. Core Test

**Observation:** Core test results confirmed that the in-situ concrete retains satisfactory compressive strength, with no major signs of structural defects.

### Conclusion and Recommendations

A detailed investigation of the RNA Mirage building—including tests such as the Rebound Hammer Test, Ultrasonic Pulse Velocity Test, Core Test, Carbonation Test, and Chloride Content Analysis—revealed significant variations in the compressive strength of concrete in both beams and columns. At certain locations, the compressive strength was found to be as low as 25 N/mm<sup>2</sup>, indicating poor quality construction practices during the original execution.

Such low compressive strength values suggest high permeability in the concrete, making it more susceptible to the ingress of harmful environmental agents such as carbon dioxide and chlorides. This accelerates the corrosion of embedded steel reinforcement and leads to the gradual disintegration of the concrete cover.

Based on the findings of this case study, the following conclusions and recommendations are proposed to ensure durable concrete construction and minimize the need for structural rehabilitation in the future:

1. **Strict Quality Control During Construction:** Proper mix design, material selection, and workmanship must be ensured to achieve the desired strength and durability of concrete.
2. **Use of Low-Permeability Concrete:** Incorporating supplementary cementitious materials and adopting mix designs that reduce permeability can significantly enhance durability.
3. **Adequate Concrete Cover:** Sufficient concrete cover over reinforcement must be maintained to protect against environmental exposure and reduce the risk of corrosion.
4. **Regular Maintenance and Monitoring:** Periodic inspections and timely maintenance interventions are essential to detect early signs of distress and prevent major structural issues.

5. **Adoption of Protective Measures:** Use of corrosion inhibitors, surface sealants, and protective coatings can help mitigate the impact of aggressive agents.
6. **Training and Supervision:** Skilled labor and on-site supervision are critical in ensuring compliance with design and construction standards.

Implementing these recommendations can significantly improve the lifespan and performance of concrete structures, reducing long-term maintenance costs and ensuring safety and stability for occupants.

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